Hochschule Fulda
University of Applied Sciences

Master Team Project Fall 2025
Team-2
FuldaNExus

Name	Email	Back end or front,End affinity/lead or GitHub master
Peddada Ramachandra Sai Darshan	ramachandra-sai-darshan.peddada@informatik.hs-fulda.de	Team Lead, QA and document master
Tanmay Manjunath	tanmay.manjunath@informatik.hs-fulda.de	Front End Lead
Sahaj Bikram Malla	sahaj bikram.malla@informatik.hs-fulda.de	Back End Lead, DB
Bikrant Bidari	bikrant.bidari@informatik.hs-fulda.de	GIT Master and Cloud
Meet Kalathiya	meet-mukeshbhai.kalathiya@informatik.hs-fulda.de	Frontend
Karan Kamath	karan-karunakar.kamath@informatik.hs-fulda.de	Backend

Milestone - 1

History Revision Table:

Version	Revised on
1.0	06.11.2025

TABLE OF CONTENTS

SR NO	TITLE	PAGE NO
1	Executive Summary	3
2	Personae and main Use Cases	5
3	List of main data items and entities	8
4	Initial list of functional requirements	13
5	List of non-functional requirements	16
6	Competitive analysis	17
7	High-level system architecture and technologies used	19
8	Team and roles	20
9	Checklist:	21

CONTENT

1.Executive Summary

Events are a major factor in the university life ecosystem and have a major impact on the connection, the learning, enrichment, and the community spirit building. However, the event management landscape is still a difficult one, even with the help of the existing commercial solutions, and in some cases the institutions will have to face the difficulties of participant safety being assured, the confusion of heavy logistics being streamlined, student engagement being maximized, and personalized event discovery being enabled. Our project, Fulda Nexus, is nothing less than a visionary next-generation event management platform that is determined to wipe out these limitations. We want to change the regular event experience to be a seamless, very engaging, and absolutely safe journey—for the organisers, the event planning will be a breeze, and for the whole university community, participation will be thrilling. By taking care of the well-being, keeping the connections deeper, and making the access easier, Fulda Nexus will definitely be a digital cornerstone for the modern university.

Fulda Nexus presents a complete and smart set of functions that are characterised by the unique specificities of a university setting. Not only are the powerful basic features like intuitive event registration, a flexible ticketing system (both free and paid), comprehensive event search, mobile access without interruption, and venue management available, but also a great number of new and innovative services, which are offered by Fulda Nexus, are introduced.

1. **Event Safety SOS Co-pilot:** This essential aspect connects a sophisticated incident management system straight onto the platform. Directly sending alerts to campus security, a panic button lets users respond fast. The system additionally enables thorough incident logging related to event metadata, therefore giving all participants and organizers an unmatched level of safety and peace of mind..

2. **Reward Engine**: In order to actively encourage participation and interaction, a very dynamic Reward system is integrated. Users receive points and rewards for participation, active involvement in activities, and good contributions. This system will encourage the users, which results in increased engagement and overall activity.

3. **Semantic/Advanced Search**: Our platform goes beyond just matching keywords, and it utilises semantic understanding together with advanced algorithms to deliver

personalised event recommendations of the highest quality. Users have the option to use Natural language queries (for instance, "workshops for aspiring entrepreneurs" or "sustainability, social events") to find events that are the best fit for their study areas, social activities, and skill improvement objectives.

4. **Secure Ticket Reselling**: Student schedules change all the time. That's why Fulda Nexus built a secure in-app marketplace where you can resell tickets you can't use. You get peace of mind knowing the handoff is safe and transparent. Other students can grab those tickets instead of missing out, so events stay full and nothing goes to waste. Plus, the whole process follows university rules.

5. **Community Chat for Events:** The platform includes a built-in communication option. Registered participants can use it to link up with one another, exchange ideas, arrange in-person or virtual discussions, and dive into talks about the related topic. In the end, this setup builds a deeper group cohesion. It also opens up better ways to network. Plus, it adds more value to the overall collaborative experience, from preparation through to the follow-up afterward.

2. PERSONA AND MAIN USE CASES

2.1 Personae

Persona 1: Lena – The Event Enthusiast

Age: 21
Role: Bachelor Student, Media and Communication Studies
Tech Skills: High
Goals: Discover and attend as many interesting university events as possible; track her participation history; earn recognition.
Pain Points: Scattered event information across multiple platforms; no unified calendar or reward system; often misses registration deadlines.

Lena loves exploring campus life and regularly attends concerts, workshops, and hackathons. She wants a single, mobile-friendly app where she can browse, register, and earn badges for participation.

Persona 2: Dr. Markus Weber – The Organizer

Age: 42
Role: Professor and Innovation Lab Coordinator
Tech Skills: Intermediate
Goals: Create and manage official events efficiently; monitor registrations; ensure participant safety during events.
Pain Points: Manual handling of registrations and attendance; no instant feedback or safety alert mechanism during events.

Dr. Weber organizes research talks and competitions. He values an intuitive interface to publish events, control capacity, and activate the SOS Co-pilot to handle any emergency quickly.

Persona 3: Sara – The International Student

Age: 24
Role: Master's Student in Global Software Development
Tech Skills: Good
Goals: Find events related to her studies and interests; connect with other students; feel secure during off-campus or late-evening events.
Pain Points: Overwhelmed by generic event listings; no clear recommendations; worries about security at night events.

Sara prefers relevant, personalized search results and wants the SOS feature integrated with campus security contacts for safety assurance.

Persona 4: Jonas – The System Administrator

Age: 29
Role: Technical Support and Infrastructure Specialist
Tech Skills: Expert
Goals: Maintain a reliable, secure system; monitor data integrity; prevent misuse of ticket reselling and student accounts.
Pain Points: Duplicate registrations; fraudulent resales; lack of visibility into technical performance metrics.

Jonas ensures the backend runs smoothly. He appreciates proper access control, event logs, and validation mechanisms for ticket transfers.

Persona 5: Emma – The Club Representative

Age: 22
Role: Event Coordinator, Student Cultural Club
Tech Skills: Above average
Goals: Promote club events effectively; engage students through gamified participation; manage attendance lists efficiently.
Pain Points: Difficult promotion; low attendance; no digital recognition for loyal participants.

Emma wants an interactive system that tracks attendance through QR check-ins, awards participation badges, and helps her visualize engagement trends.

2.2 Use Cases

Use Case 01 – Smart Event Discovery

Primary Actor: Lena (Student)
Scenario:
Lena opens *FuldaNexus* and searches “tech workshops this weekend.”
The **semantic search engine** interprets her intent and shows events that match her academic background and previous interests.
She filters for “free events” and “in-person only,” adds two to her favorites, and receives reminders before they begin.

Use Case 02 – Event Creation with Safety Integration

Primary Actor: Dr. Markus Weber (Organizer)
Scenario:
Dr. Weber logs into the *Organizer Dashboard*, fills in details (event title, date, venue, max participants), uploads a banner, and enables **Safety SOS Co-pilot**.
If any participant triggers an SOS alert during the event, it sends a real-time notification with location details to the organizer and campus security.
The event concludes with auto-generated attendance data.

Use Case 03 – Gamified Engagement System

Primary Actor: Emma (Club Representative)
Scenario:
Emma organizes a cultural fest and activates **Reward Mode**.
Participants earn points for early registration, volunteering, and attendance.
The leaderboard displays top students by accumulated points.
After the event, badges are automatically added to each participant's *FuldaNexus* profile, motivating continued participation.

Use Case 04 – Ticket Reselling / Transfer

Primary Actor: Sara (International Student)
Scenario:
Sara buys a paid ticket for a movie night but later can't attend.
Through *FuldaNexus*, she opens "My Tickets" → "Resell."
The system validates the ticket, lists it on the internal marketplace, and securely transfers it to another student once sold.
The resale is tracked to maintain transparency and avoid duplication.

Use Case 05 – Admin Oversight and System Logs

Primary Actor: Jonas (System Administrator)
Scenario:
Jonas accesses the admin panel to review ticket transfers, event creation logs, and attendance reports.
He detects a user attempting multiple ticket resales and restricts further listings.
He exports an activity summary for system auditing and verifies that the SOS module has no false alerts.

3. LIST OF MAIN DATA ITEMS AND ENTITIES

This section defines all the system's key data elements, users, and entities. Each item includes its name, meaning, and intended use. These terms will remain consistent across future documentation, UI design, and software development.

3.1 User Types (Entities with Behavior and Privileges)

1. Registered User

Meaning: Verified Fulda University student or staff member with an email ending in `@hs-fulda.de`.
Usage: Primary user who can create, browse, and join events, interact through event chats, and manage personal participation.
Privileges:

- Create, edit, or delete their own events
- Browse, search, and filter events
- Register for and cancel participation in events
- Participate in event group chats
- Edit personal profile and upload media
- Report inappropriate content or users

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2. Guest

Meaning: Non-registered or logged-out visitor to the platform.
Usage: Can view public event listings and details but cannot interact or register.
Privileges:

- Browse upcoming events and view limited event information
- Search by category or keyword
- No ability to create events, chat, or register

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3. Administrator

Meaning: Authorized Fulda Events platform manager responsible for reviewing, approving, and moderating user-generated content.
Usage: Ensures compliance with university guidelines, handles reports, and maintains platform integrity.
Privileges:

- Approve or reject newly created events
- Edit or delete any event or message violating policy
- Suspend or deactivate user accounts
- Access and review moderation logs
- Manage categories and site-wide content

3.2 Core Data Entities and Structures

1. UserProfile

Meaning: Stores account and personal details of a registered user.
Fields: UserID, FullName, Email, Password (hashed), Role (User/Administrator), ProfilePictureURL, Bio, Status, JoinedDate.
Usage: Represents individual user identity and manages access levels and personalization.

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2. Event

Meaning: Represents a specific activity organized within or near Fulda (e.g., academic workshop, cultural event, sports meetup).
Fields: EventID, Title, Description, OrganizerID, CategoryID, Date, StartTime, EndTime, Location, MediaURLs, ApprovalStatus, CreatedAt, UpdatedAt.
Usage: Central object connecting users, participation records, and chats. Viewable through event listings and detail pages.

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3. EventParticipation

Meaning: Represents a user's registration and participation status in an event.
Fields: ParticipationID, EventID, UserID, RegistrationDate, Status (Registered/Cancelled/Attended).
Usage: Tracks attendance and grants access to the event's chat and notifications.

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4. Category

Meaning: Defines types of events to improve discoverability and organization.
Fields: CategoryID, Name, Description.
Usage: Enables users to browse or filter events based on themes such as Sports, Academic, Cultural, Volunteer, etc.

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5. Chat

Meaning: Represents the real-time discussion area associated with an approved event.
Fields: ChatID, EventID, CreatedAt.
Usage: Automatically created when an event is approved. Only accessible to registered participants of that event.

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6. Message

Meaning: Individual communication sent within an event chat.
Fields: MessageID, ChatID, SenderID, Content, Timestamp, IsEdited, IsDeleted.
Usage: Enables communication between participants and organizers for coordination or questions.

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7. Media

Meaning: Represents media items (images, flyers, videos) uploaded for events or user profiles.
Fields: MediaID, UploadedByUserID, AssociatedEventID, FileURL, MediaType (Image/Video/Link), UploadedAt.
Usage: Enhances event descriptions and user profiles with visual content.

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8. Notification

Meaning: Alerts users about important actions related to events or participation.
Fields: NotificationID, RecipientID, Type (System/Event/Chat), Message, Status (Read/Unread), Timestamp.
Usage: Keeps users informed about event approvals, new messages, or updates.

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9. Report

Meaning: Used by users to report inappropriate content or activities.
Fields: ReportID, ReporterID, TargetType (User/Event/Message), Reason, Timestamp, Status.
Usage: Sent to administrators for moderation and action.

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10. AuditLog

Meaning: Records all administrative actions and moderation activities for accountability.
Fields: LogID, AdminID, ActionType, TargetEntityID, EntityType, Timestamp, Notes.
Usage: Provides traceability and ensures compliance with moderation policies.

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11. Feedback (Optional Extension)

Meaning: Allows users to give feedback about attended events.
Fields: FeedbackID, EventID, UserID, Rating, Comment, Timestamp.
Usage: Builds trust and helps organizers improve event quality.

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12. Rewards

Meaning: Tracks user engagement and rewards through credits earned for platform activity.
Fields: CreditID, UserID, TotalCredits, EarnedFrom (ActivityType), RedeemedCredits, RedemptionDate, Level (Bronze/Silver/Gold).
Usage: Encourages consistent user engagement; credits can be redeemed for discounts on paid event tickets.
Rules:

- Credits are earned by joining or creating events, or engaging in chats.
- Credits cannot be exchanged for money.
- Credit balance and level are shown in the user's profile.

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13. Ticket & Reselling

Meaning: Represents tickets purchased by users for specific events, including resale capabilities.
Fields: TicketID, EventID, BuyerID, PurchaseDate, Price, Status (Active/Cancelled/Resold/Used), IsRefundable, ResellEligible, ResoldToUserID, OriginalPrice.
Usage: Tracks ticket ownership, supports reselling, and ensures compliance with resale rules.
Rules:

1. A user may purchase **up to 4 tickets per event**.
2. Events marked **non-refundable** may allow **reselling** if the organizer permits.
3. Reselling is only allowed **on the platform** and **at or below the original purchase price (MRP)**.
4. Organizers decide at event creation whether **reselling** is permitted.
5. Tickets cannot be resold after being marked as “Used” or after the event date.

3.3 Relationships Overview (Logical View)

- A **UserProfile** can **organize** multiple **Events**.
- A **UserProfile** can **participate** in multiple **Events** via **EventParticipation**.
- Each **Event** belongs to one **Category**.
- Each **Event** has one **Chat**.
- Each **Chat** contains multiple **Messages**.
- **Media** may belong to either a **UserProfile** or an **Event**.
- **GamificationCredits** belong to a **UserProfile** and are earned through activities.
- **Tickets** link **Users** and **Events**, with enforced purchase/resell limits.
- **Administrators** manage **Reports**, **AuditLogs**, and **Event approvals**.

4. INITIAL LIST OF FUNCTIONAL REQUIREMENTS

1. User Management & Authentication

- The system shall allow users (attendees, organizers, administrators) to register for an account using their university credentials or create a new account.
- The system shall allow users to log in securely with their credentials.
- The system shall support different user roles (e.g., Attendee, Event Organizer, Delivery Agent, Administrator), each with distinct permissions.
- The system shall allow users to view and edit their profile information (e.g., name, contact details, profile picture, interests).
- The system shall allow users to reset their password.

2. Event Creation & Management (Organizer-Specific)

- The system shall allow authorized Event Organizers to create new event listings, including event name, description, date(s), time(s), location (physical/virtual), capacity, and category.
- The system shall allow Event Organizers to specify ticket types (e.g., free, paid, VIP) and their respective prices, quantities, and availability periods.
- The system shall allow Event Organizers to upload event-related media (images, videos).
- The system shall allow Event Organizers to view, edit, or cancel their own events.
- The system shall provide Event Organizers with a dashboard to view event analytics (e.g., registration numbers, attendance, ticket sales).
- The system shall allow Event Organizers to manage attendee lists, including checking in attendees.

3. Event Discovery & Search

- The system shall display a comprehensive list of upcoming and past events.
- The system shall allow users to search for events by keywords, categories, dates, and locations.
- The system shall provide advanced filtering options (e.g., free events, specific departments, event type).
- (Semantic/Advanced Search) The system shall allow users to perform natural language searches and receive relevant event recommendations based on their profile interests, past attendance, and semantic understanding of event content (e.g., "workshops to improve my public speaking skills").
- The system shall display detailed information for each event, including description, schedule, location map (if physical), and organizer contact.
- The system shall allow users to add events to a personal "favorites" or "watch" list

4. Registration & Ticketing

- The system shall allow users to register for free events.
- The system shall allow users to purchase tickets for paid events.
- The system shall generate unique digital tickets/QR codes for each successful registration/purchase.
- The system shall provide users with a record of their registered/purchased tickets within their profile.

6. Event Safety SOS Co-pilot (USP)

- The system shall provide an easily accessible "SOS" or panic button within the event interface for attendees.
- Upon activation of the SOS button, the system shall immediately alert designated campus security personnel with the user's location (if permissions granted and available) and event context.
- The system shall allow security personnel/administrators to log incident details tied to specific events or attendees.
- The system shall provide real-time communication channels between the SOS activator and security personnel.

7. Rewards (USP)

- The system shall track user actions relevant to gamification (e.g., event attendance, successful interactions, positive reviews).
- The system shall award points and badges to users based on predefined criteria.
- The system shall display a user's earned badges and points on their profile.
- The system shall include a leaderboard displaying top-ranking users based on accumulated points/achievements.
- The system shall allow for the definition and redemption of rewards associated with gamification achievements (e.g., exclusive access, discounts).

8. Reselling of Tickets (USP)

- The system shall allow users who have purchased a ticket to list it for resale.
- The system shall provide a dedicated interface for browsing tickets available for resale.
- The system shall facilitate a secure transfer of ownership of a resold ticket from the original owner to the new buyer.
- The system shall enforce university-defined policies regarding ticket resale (e.g., price caps, transfer limits).

9. Community Chat for Events (USP)

- The system shall provide a dedicated chat room for each event, accessible only to registered attendees of that event.
- Users shall be able to send and receive text messages within event chat rooms.
- The system shall allow users to view other participants in the event chat.

- The system shall support basic moderation capabilities for chat rooms (e.g., reporting inappropriate content).

5. LIST OF NON-FUNCTIONAL REQUIREMENTS

1. **Tools and Deployment:** The application must be developed, tested, and deployed using tools and servers approved by the Class CTO and agreed upon in Milestone 0. Application delivery must be from the chosen cloud server.
2. **Browser Compatibility:** The application must be optimized for standard desktop/laptop browsers, rendering correctly on the two latest versions of two major browsers.
3. **Mobile Rendering:** All or selected application functions are required to render well on mobile devices. (The project prioritizes desktop browsers. but must use a responsive UI).
4. **Data Storage:** Data must be stored in the database located on the team's deployment cloud server.
5. **Concurrency:** No more than 50 concurrent users shall be accessing the application at any time.
6. **Privacy:** The privacy of users must be protected., and all privacy policies must be appropriately communicated to the users.
7. **Language:** The language used must be English; no localization is needed.
8. **Usability:** The application must be very easy to use and intuitive.
9. **Architecture:** The application should follow established architecture patterns.
10. **Maintainability:** The application code and its repository must be easy to inspect and maintain.
11. **Analytics:** Google Analytics shall be used (this is optional for Fulda teams).
12. **Email Constraint:** No email clients are allowed.
13. **Payment Constraint:** Payment functionality, if any, shall not be implemented or simulated in the UI.
14. **Security:** Basic best practices for site security (as covered in the class) must be applied for main data items.
15. **Media:** The application must be media-rich (images, video, etc. using standard market formats.
16. **Processes:** Modern Software Engineering (SE) processes and practices must be used as specified in the class, including collaborative and continuous software development.
17. **Code Management:** Each team must use their own GitHub repository (set up by instructors) for code development, management, and required documentation such as formal milestones.
18. **Display Text:** The application UI must prominently display the following exact text on all pages at the top: "Fulda University of Applied Sciences Software Engineering Project, Summer 2025 For Demonstration Only"

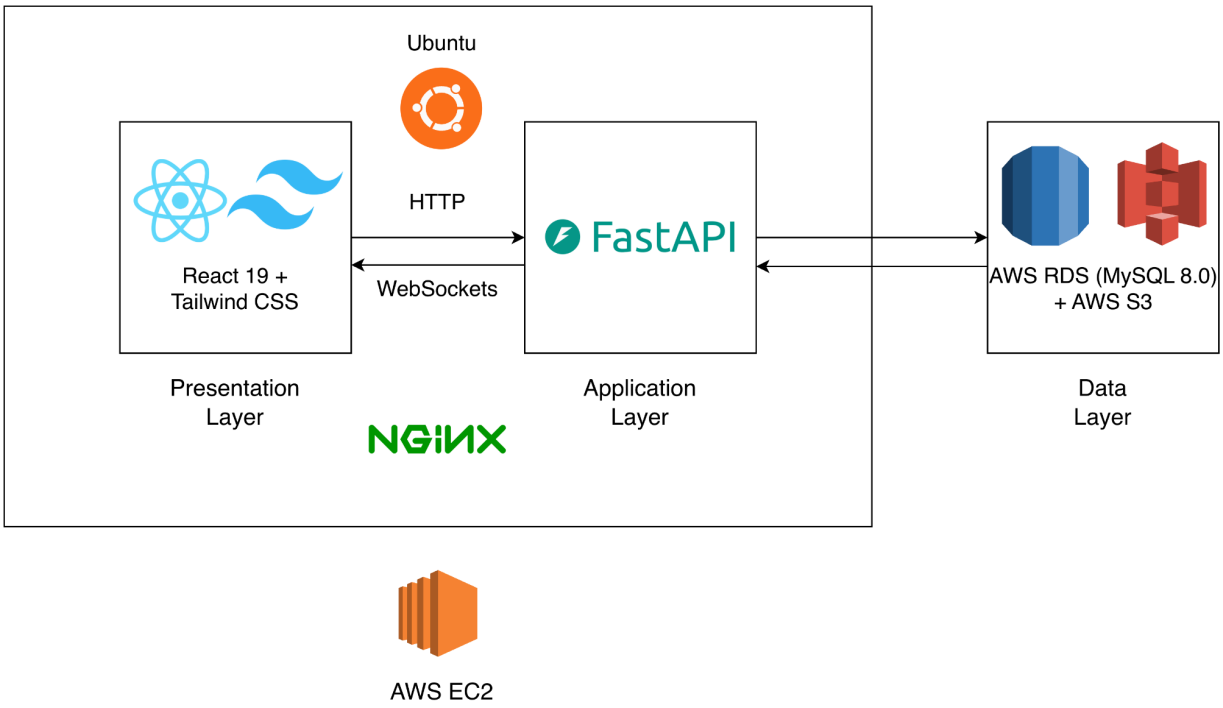
6. COMPETITIVE ANALYSIS

Features	Ticket Master	Event Brite	Cvent	vFairs	Bookmyshow	FuldaNexus
Registration / Ticketing (Free/Paid)	Yes	Yes	Yes	Yes	Yes	**Yes**
Search / Event Discovery	Yes	Yes	Yes	Yes	Yes	**Yes**
Venue Managementent / Seating	Yes	No	Yes	No	Yes	**Yes**
Mobile Support	Yes	Yes	Yes	Yes	Yes	**Yes**
Hybrid/Virtual Event Support	No	No	Yes	Yes	No	**Yes**
Event Safety SOS Co-pilot	No	No	No	No	No	**Yes (USP)**
Rewards System	No	No	No	No	No	**Yes (USP)**
Semantic/ Advanced Search	No	Parti…	No	No	Parti…	**Yes (USP)**
Secure Ticket Reselling	Yes	No	No	No	No	**Yes (USP)**

Event Specific Chat/ Community chat	No	No	No	No	No	**Yes**

7. HIGH-LEVEL SYSTEM ARCHITECTURE AND TECHNOLOGIES USED

1. Server Host: AWS (Cloud Infrastructure)
2. Operating System: Ubuntu 24.04 LTS
3. Web Server: NGINX 1.28.0
4. Server-Side Language: Python 3
5. Backend Framework: FastAPI (Python)
6. Database: MySQL 8.0
7. Frontend Framework: React 19
8. Styling Framework: Tailwind CSS 3.2.x
9. Real-Time Communication: WebSockets (for chat and live updates)
10. IDE: Visual Studio Code 1.79.x
11. Version Control & CI/CD: GitHub and GitHub Actions
12. Project Management: GitHub Projects
13. Team Communication: WhatsApp



8. TEAM AND ROLES

Name	**Email**	**Back end or front,End affinity/lead or GitHub master**
Peddada Ramachandra Sai Darshan	ramachandra-sai-darshan.peddada@informatik.hs-fulda.de	Team Lead, QA and document master
Tanmay Manjunath	tanmay.manjunath@informatik.hs-fulda.de	Front End Lead
Sahaj Bikram Malla	sahaj bikram.malla@informatik.hs-fulda.de	Back End Lead, DB
Bikrant Bidari	bikrant.bidari@informatik.hs-fulda.de	GIT Master and Cloud
Meet Kalathiya	meet-mukeshbhai.kalathiya@informatik.hs-fulda.de	Frontend
Karan Kamath	karan-karunakar.kamath@informatik.hs-fulda.de	Backend

9. CHECKLIST

Item	Status
Team found a time slot to meet (online) outside of the class	Done
GitHub master chosen	Done
Team decided and agreed together on using the listed SW tools and deployment server	Done
Team ready and able to use the chosen back and front end frameworks and those who need to learn are working on learning and practicing	On Track
Team lead ensured that all team members read the final M1 and agree/understand it before submission	Done
GitHub organized as discussed in class (e.g. master branch, development branch, folder for milestone documents etc.)	On Track